MAPS FROM REDUCED SCHEMES INTO A CLOSED SUBSCHEME ARE DETECTED ON POINTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2495-2507, following the statement at 2485-2493.

Editorial note. The proof covers the nontrivial direction; the converse is a one-line set-theoretic triviality and the closing 'In particular' applies the proven iff with Z = X_red.

Allowed standard prerequisites (editorial).

- Reduced rings have zero nilradical. lemma-reduced (same chapter, cited at ~2503) with Algebra, Lemma algebra-lemma-Zariski-topology (cross-chapter, cited at ~2504). b in the intersection of all primes implies b = 0.
- Factoring through a closed subspace. lemma-characterize-closed-subspace (card L26388, cited at ~2506). Vanishing of the ideal pullback gives the factorization.
- Reduced induced closed subscheme. lemma-reduced-closed-subscheme (card L26385; used for the closing remark). X_red exists with underlying space X.

Source reference. Original reference 00E0: focused-seeds/geometry/S9-schemes-map-into-reduction/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HP: sources/stacks-schemes/pinned-original-source.tex, source lines 632–632. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J1: sources/stacks-schemes/pinned-original-source.tex, source lines 2371–2371. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $Z \subset X$ be a closed subscheme. Let Y be a reduced scheme. A morphism $f : Y \to X$ factors through Z if and only if $f(Y) \subset Z$ (set theoretically). In particular, any morphism $Y \to X$ factors as $Y \to X_{red} \to X$.*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume $f(Y) \subset Z$ (set theoretically). Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal sheaf of Z. For any affine opens $U \subset X$, $\mathrm{Spec}(B) = V \subset Y$ with $f(V) \subset U$ and any $g \in \mathcal{I}(U)$ the pullback $b = f^\sharp(g) \in \Gamma(V, \mathcal{O}_Y) = B$ maps to zero in the residue field of any $y \in V$. In other words $b \in \bigcap_{\mathfrak{p} \subset B} \mathfrak{p}$. This implies $b = 0$ as B is reduced (Lemma 01J1, and Algebra, Lemma 00E0). Hence f factors through Z by Lemma 01HP. $\square$